Everything proposed, on one page: who touches it, what runs it, what it talks to.

Read top to bottom. People, then the agent that operates each service, then the service itself, then the standard interfaces, then the systems the center buys. Built things are solid and warm. Bought things are dashed. Nothing crosses the clinical wall on the right.

Prepared for **Dr. Luis Ramos**, **Fernando Escarzaga**, and **Jesse Martinez**
by **Richard Finlay**, RGF Global Ventures LLC

PEOPLE

- **Rick, operator** — Builds, runs, supervises. Access revocable.
- **Prior auth staff** — Signs every submission and every appeal.
- **Owner and partners** — Ask the owner agent. Approve any sign-off change.
- **Schedulers, front desk** — Scheduling, intake, waitlist. Approve backfill offers.
- **Patients** — Reminders, offers, records requests, inbound calls.
- **Referring physicians** — Orders in, documentation, reports out.
- **Billing staff** — Claims. Their vendor exposes the eligibility feed.
- **Radiologists, readers** — Clinical decisions are theirs alone.

AGENTS

- **Prior auth agent** — Tier B. Reads referrals, flags gaps, drafts appeals. Acts only via the gate.
- **Owner agent** — Tier A. Read-only questions over the center's data. Never acts.
- **Backfill agent** — Tier B. Proposes who gets a freed slot. A human approves.
- **Records agent** — Tier B. Classifies requests, drafts replies. Identity check is human.
- **Voice agent** — Tier C. The vendor's own. Rick specifies, integrates, supervises.
- **No agent. Plain software, no model involved.** — Reminders, intake, report delivery and billing run on the vendors' own products as bought.

HUMAN SIGNS · HUMAN APPROVES · HUMAN VERIFIES

SERVICES

- #1 #2 BUILD — **Prior auth and eligibility engine** — Order detection. Reads the billing vendor's 270/271 feed. Packet assembly, submission, deadline tracking, appeal queue, weekly rates.
- #4 BUILD — **Owner reporting and reader reconciliation** — Joins RIS data to billing data. Monday report, appeal rate as headline. Reader paid-versus-read view.
- #5 #10 BUILD, SMALL — **Backfill and waitlist** — Cancellation in, ranked offer list out. One per-patient rate governor across every sender.
- #6 #7 BUILD, SELECTIVELY — **Records in, prior studies out** — Intake, 30-day clock from receipt, fee rules, DICOM retrieve. Outbound prior-study tracker.
- #9 BUY, I INTEGRATE — **Voice vendor product** — Vendor under its own BAA. Spanish evaluation set. Human escalation on any symptom language.
- #8 #10 BUY, THE RIS — **Reminders, self-scheduling, intake, notices** — RIS native. Verify TCPA caps, Spanish, and events exposed to backfill.
- #12 BUY, RIS AND PACS — **Referrer communication and report delivery** — Standard RIS and PACS function. Referrer volume feeds reporting.
- BUY — **Billing, claims, eligibility feed** — Clearinghouse function. Exposes EDI 270/271. Not rebuilt. The engine reads the feed and acts on it.

THE FDA LINE. NEVER CROSSED.

- #13 DECLINED — ~~Anything touching the image~~ — The FDA line. No image interpretation. No protocol selection. No study recommendation. No appropriateness judgment.

  Software on this side is a device unless it has an FDA pathway. Everything left of the wall stays inside the administrative support exclusion, 21 U.S.C. 360j(o)(1)(A).

Standard interfaces only, secured by four contract terms before signature (FREE): named API access to the center's own data, HL7 interface count and fee schedule fixed, data export rights at no charge, DICOM query permissions.

INTERFACES

- **HL7 v2 ORM, SIU** — EDI 270/271 read; EDI 278; Payer FHIR PA API; Human submission path
- **Data exports** — RIS report export or database; Billing system export
- **HL7 v2 SIU** — Cancellation events; Consent and contact preferences from intake
- **DICOM C-FIND, C-MOVE** — DICOM import; Fax, web form, portal intake
- **RIS scheduling API** — Vendor books against it; Human escalation path
- **RIS native** — Reminder events out to the backfill service; One shared rate cap
- **RIS and PACS native** — Orders in, reports out; Referrer volume to reporting
- **EDI 270/271** — Exposed by the billing vendor; Claims stay with the vendor

- #11 DECLINED, PHASE ONE — ~~Balance-due reminders~~ — Financial content is outside the TCPA exemption. Needs its own consent basis.

SYSTEMS

- **RIS** — Orders, schedule, cancellations, reminders. Chosen by the center.
- **PACS** — Images. DICOM query and retrieve for authorized systems.
- **Billing vendor, clearinghouse** — Claims. Exposes EDI 270/271 and, where supported, 278.
- **Payers and RBMs** — FHIR PA API from 2027-01-01, EDI 278, or a human-submitted packet.
- **Voice vendor** — Own BAA. Telephony and transcripts stored in the US.
- **Prior facilities, El Paso** — Hospitals and imaging groups. Fax or portal out, DICOM back.
- **Referring practices** — Orders and clinical documentation in. Reports out.
- **Model provider** — Named vendor, own BAA, US residency. Replaceable.

- #3 FREE — **Contract terms** — The precondition for every interface here. Free only before signature.

BAA before any PHI flows. 45 CFR 164.502(e). Center, model provider, voice vendor, each under its own agreement.

BAA envelope gap. The AI vendor's agreement excludes Computer Use and Web Fetch, so payer-portal screen-driving against PHI is not available. Design goes through APIs, EDI, and a human step.

Texas SB 1188. Texas patient data stored in the US as of 2026-01-01. Applies to every component on this page, including the model provider.

Two deliverables per build. A deterministic service that runs with no model, plus an agent surface that operates it. The agent is the replaceable part.

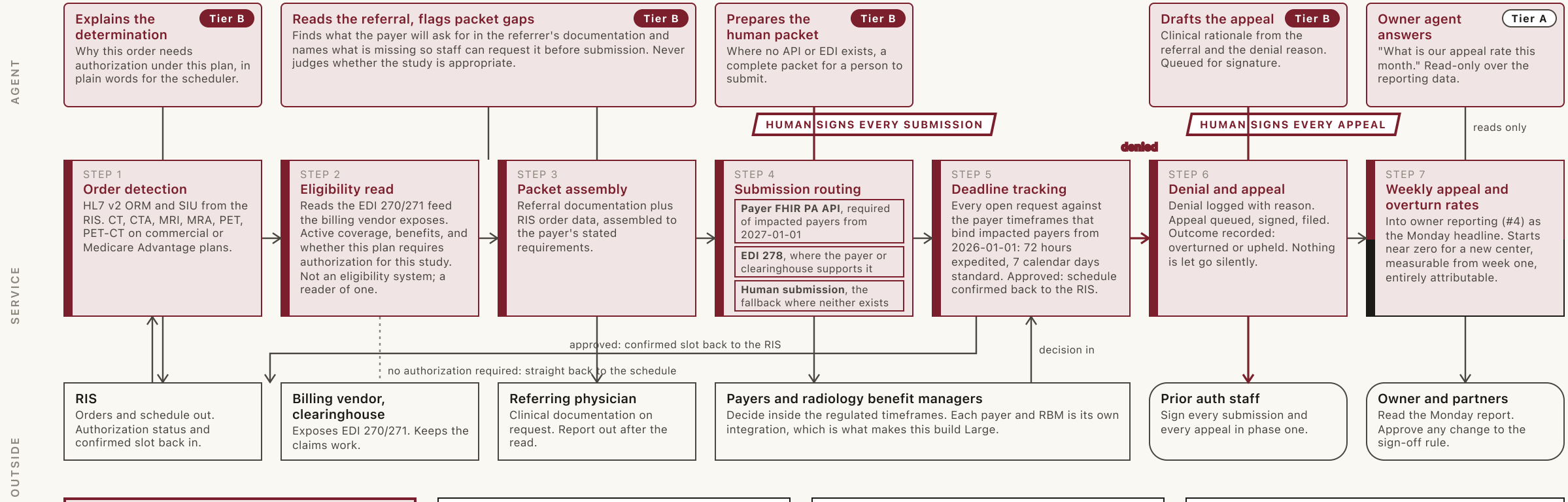
LEGEND — Tier A: owner agent, read-only · Tier B: workflow agent, acts, a human signs · Tier C: patient-facing voice, bought · No agent: plain software · Build · Buy · Free · Declined

Rick's engine, used by the center under the contract · Center's outright · HUMAN SIGNS: signature gate, every item until the owner approves a change

Prior authorization: one pipeline, three submission paths, a human at every gate.

Left to right is the life of one advanced imaging order. The upper lane is the agent surface. The lower lane is the deterministic service that runs with no model at all. The agent never reaches the world except through a signature.

Candidates **#1** and **#2**
Tier **B**, **Build**, Rick's engine, terms in the contract



CONSTRAINT, THE MOST FRAGILE ITEM

No portal screen-driving against PHI.

The AI vendor's BAA excludes Computer Use and Web Fetch, the two capabilities one would use to drive a payer portal with no API. So the third path is a human, by design. A locally hosted model is a specification-phase option for that gap. It is not promised.

GRADUATION CRITERION, WRITTEN IN THE SPECIFICATION

Every item, until the owner says otherwise.

Sign-off moves from every item to a sample only after a defined volume at a defined accuracy, measured, and only with the owner approving the change. Until then, every submission and every appeal carries a human signature.

THE FDA LINE, INSIDE THIS PIPELINE

Claims, billing, eligibility. Never appropriateness.

"Claims or billing information" and "determination of health benefit eligibility" are named in the administrative support exclusion, 21 U.S.C. 360j(o)(1)(A). The agent never recommends which study to order or whether a study is clinically appropriate. That would leave the exclusion.

TWO DELIVERABLES

The service is the capability. The agent is the operator.

Steps 1 through 7 run with no model: auditable, testable, in the specification, and what any integrator can take over. The agent lane is the replaceable part. Model vendor, BAA envelope, and pricing can all change without breaking the center.

Dependencies named in the inventory: RIS order and scheduling feed as a contract term; payer FHIR Prior Authorization APIs under CMS-0057-F, which binds payers and not providers; EDI 278 or human submission for plans outside the rule. El Paso payer mix is not verified and nothing here forecasts recovered dollars.

Who touches what, and what each person is allowed to do.

A row is a person. A column is a service. The signature marks are the places the system stops and waits for a human. Radiologists appear once, to show where the system is not.

Access is role-based and logged
Rick's own access is **revocable by the center**

PERSON	PRIOR AUTH AND ELIGIBILITY · TIER B	OWNER REPORTING AND OWNER AGENT · TIER A	BACKFILL AND WAITLIST · TIER B	RECORDS IN, PRIOR STUDIES OUT · TIER B	PATIENT VOICE, VENDOR · TIER C	RIS REMINDERS, SELF-SCHEDULING, INTAKE · NO AGENT
Owner and partners Dr. Ramos, Mr. Escarzaga, Mr. Martinez	See every open request, deadline, denial, and appeal outcome. Approve the graduation criterion that changes the sign-off rule. APPROVES GRADUATION	Read the Monday report. Ask the owner agent open questions: volumes, cancellations, no-shows, payer mix, appeal rate, reader turnaround. Define what the one page contains.	See fill rate and the ranking rules. Change the ranking policy.	See the request log, open clocks, and fees charged.	Hear call recordings and transcripts. Set the escalation rules. Approve the Spanish evaluation bar.	Configure through the RIS vendor. Confirm caps and Spanish at selection.
Prior auth staff The signature	Review every assembled packet and every drafted appeal. Submit by hand on the human path. Nothing goes to a payer without their signature. SIGNS EVERY SUBMISSION SIGNS EVERY APPEAL	Read-only on their own queue metrics.	No access.	No access.	No access.	No access.
Schedulers, front desk	See authorization status on each order so a scan is not booked before it is cleared. Request missing documentation from the referrer.	No access.	Receive the ranked offer list on a cancellation and approve it before anyone is contacted. Maintain consent and contact preferences at intake. APPROVES THE OFFER LIST	Log a walk-in or phoned request so the 30-day clock starts at receipt.	Take escalated calls, including any call with symptom language. Book what the vendor could not.	Operate the RIS. Intake forms and reminders run as the vendor ships them.
Billing staff	See authorization numbers attached to orders for claims. The eligibility feed is their vendor's; the engine only reads it.	Provide the billing export the reporting joins depend on.	No access.	Set the fee schedule inside the permitted components. Retrieval labor is never charged.	No access.	No access.
Radiologists, readers Including Dr. Ramos as a 1099 reader	Untouched. The system never recommends a study or judges appropriateness.	Reader reconciliation view: what was read against what was paid.	Untouched.	Ask for a prior study; the tracker chases it and imports what returns. The read is theirs.	Untouched.	Untouched.
Referring physicians	Orders arrive from them through the RIS. Documentation requests go to them. They are never asked to change an order.	No access. Their volume appears in the owner's view.	No access.	Can request records for their patient; same intake, same clock.	No access.	Receive reports through the RIS and PACS as bought.
Patients	No direct surface. They experience a faster clearance, nothing else.	No access.	Receive an offer for a freed slot, only to the number they gave, in Spanish or English, with opt-out in every message, inside one message a day and three a week across every sender.	Request their own records. Identity verified by a person before anything is released. HUMAN VERIFIES IDENTITY	Call in. Scheduling, directions, prep, status. Told at the start that it is an automated system. Handed to a person on any symptom language.	Reminders, self-scheduling, intake forms, results-ready notices from the RIS.
Rick, operator RGF Global Ventures	Builds and operates the four built services and their agent surfaces. Sees logs and metrics. Cannot sign a submission, approve an offer list, or release a record; those are the center's people. Access granted and revocable by the center; repository under the center's organization.				Specifies, evaluates, integrates, supervises. The contract is in the center's name.	Advises at vendor selection. No operational role.

What the human marks mean. Where a service acts on the world, a named person at the center signs first. That is design, not cost, and it is also the FDA boundary: no software here makes or recommends a clinical decision.

Backfill, records, prior studies, owner reporting: each is a small deterministic service with a bounded agent on top.

Each panel reads left to right
Human gates drawn where they exist

#5 AND #10. BACKFILL AND WAITLIST. TIER B. BUILD, SMALL. RICK'S ENGINE.

Backfill agent — Tier B
Explains the ranking and drafts the offer wording in Spanish or English.

SCHEDULER APPROVES THE LIST

RIS
Cancellation event, HL7 v2 SIU or API.

→ **Waitlist ranking**
Which waitlisted patients, in what order. The custom logic.

→ **Rate governor**
One per-patient cap across every sender: one a day, three a week, voice and text combined.

→ **Patient**
Offer to the number they provided. Opt-out in every message, honored at once.

RIS reminders (#8)
Bought. Its sends count against the same cap.

RIS
Slot filled, schedule updated, lead time shortened.

Constraints
TCPA healthcare exemption, 47 CFR 64.1200(a)(9)(iv). No financial content. Spanish is a requirement in El Paso. "Appointment schedules" is named in the FDA exclusion.

#6. INBOUND RECORDS REQUESTS. TIER B. BUILD, SELECTIVELY. CENTER'S CODE.

Records agent — Tier B
Classifies the request, drafts the response and any extension notice.

HUMAN VERIFIES IDENTITY

Intake channel
Fax, web form, portal. Patient, attorney, insurer, provider.

→ **Log and clock**
30 days from receipt. One 30-day extension, written notice inside the first 30. Outer limit 60.

→ **Retrieve**
DICOM C-FIND and C-MOVE against the PACS, the most standardized surface available.

→ **Fee and deliver**
Permitted components only: copying labor, supplies, postage. Retrieval labor is not chargeable.

PACS
Query permission is a contract term.

Status reporting
Open requests and days remaining, into the owner's view.

Why the clock is the first feature
The failure mode is a missed statutory deadline and an OCR complaint. 45 CFR 164.524 in full. Texas Chapter 181 deadline verified before the specification is final.

#7. PRIOR-STUDY RETRIEVAL. NO AGENT. BUILD, SELECTIVELY. CENTER'S CODE.

Radiologist or RIS order
A comparison study is needed. A new center opens with zero priors on file.

→ **Facility directory**
Which El Paso hospitals and groups accept which request channel. Built by hand in the first months.

→ **Request tracker**
Generates the request, tracks it, chases it.

→ **Prior facility**
Fax or portal request out. CD, portal link, or DICOM back.

PACS
DICOM import of what returns.

Said plainly
There is no network to query. DICOM exchange is not a TEFCA requirement today. The tracker is the buildable part; the network is not. Outbound requests carry PHI, so the BAA and residency rules apply.

#4. OWNER REPORTING. TIER A. BUILD. RICK'S CORE, THE CENTER'S REPORTS.

RIS export

Billing export

Prior auth engine

→ **Reporting core and joins**
RIS data joined to billing data. Volume, turnaround, referrer, revenue per slot, appeal rate, cancellations, no-shows. Reader paid-versus-read reconciliation.

→ **Monday report**
One page. Appeal rate is the headline. The owner defines the page.

→ **Owner and partners**
Read the page. Ask the agent. It sees nothing the owner does not already see, acts on nothing, never addresses a patient's care. "Business analytics" is named in the FDA exclusion.

Owner agent — Tier A
Read-only questions, plain answers. Text or call. Specified in design:

Why the owner agent is the lowest-risk agent in the inventory
Read-only. No action on the world. No patient contact. No clinical surface. Because it reads the center's own database, the AI vendor's Computer Use and Web Fetch exclusions do not apply to it. PHI is present, so the BAA and SB 1188 residency still do. Strongest candidate for the first demonstration.

Every panel follows the same standard: the deterministic service is the specification and what any integrator can take over; the agent is the replaceable operator. Regulatory figures are the regulation's own. No cost or volume benchmark is quoted because none was verified.

Who owns each layer, and the lines the system does not cross.

The left half is a commercial boundary. The right half is a regulatory one. Both are drawn so an engineer who has never met Rick can read them.

Ownership decided **in three tiers**
Boundaries drawn from the statute and the vendor's BAA

OWNERSHIP OVERLAY. MATCH THE STRIPE COLORS TO THE SERVICE BOXES ON PAGES 1 TO 4.

TIER 1

The specification. The center's, outright.

Written for an engineer who has never met Rick. Covers every service on page 1: order detection, packet assembly, deadline tracking, waitlist ranking, the records clock, the reporting joins, and the vendor evaluation criteria for the RIS, PACS, billing, and voice selections. If the partners open a second or third site, the specification is written once and deployed again, one instance per site (page 6).

TIER 2

Center-specific code. The center's, outright.

Configuration, workflows, integrations, data mappings, and reports. In a repository under the center's own organization, with Rick's access revocable. On the diagrams, this is the black stripe:

- Records intake, clock, fee rules, status reporting (#6)
- Prior-study tracker and the El Paso facility directory (#7)
- The Monday report and every center-defined view (#4)
- Every HL7, DICOM, and payer mapping, and every RIS and billing configuration

TIER 3

Rick's existing product, extended for the center. Terms in the contract.

Who owns which pieces is settled in the contract; the center keeps running all of it. On the diagrams, this is the red stripe:

- The prior authorization engine (#1, #2)
- The backfill engine (#5)
- The reporting core (#4)

Boxes with a split stripe hold Rick's engine beneath the center's configuration and mappings.

NOT OWNED BY ANYONE HERE

The RIS, PACS, billing, and voice vendor contracts are in the center's name, never routed through Rick. The model provider is a named vendor under its own BAA, chosen so it can be replaced without touching the services.

COMPLIANCE BOUNDARIES. DRAWN AS WALLS, NOT FOOTNOTES.

THE FDA LINE

Nothing interprets an image, selects a protocol, or recommends a study.

Administrative support software is excluded from the device definition at 21 U.S.C. 360j(o)(1)(A): appointment schedules, claims or billing information, business analytics, practice and inventory management, determination of health benefit eligibility. Every service here lives inside that list. The moment software touches image data or clinical appropriateness it leaves the exclusion and enters Clinical Decision Support territory at 360j(o)(1)(E). Candidate #13 is declined for that reason, out loud.

THE BAA ENVELOPE

What is inside, what is not.

A business associate agreement with the center before any PHI flows, 45 CFR 164.502(e). The AI vendor's agreement covers Skills, Web Search, Memory, Bash tool, and Text Editor tool. It **excludes Computer Use and Web Fetch**, so payer-portal screen-driving against PHI is not available. Confirm the applicable BAA version in writing.

TEXAS SB 1188

US data residency, every component.

Electronic health records with Texas patient data must be physically stored in the US as of 2026-01-01. This binds the RIS, the PACS, the billing vendor, the voice vendor's telephony and transcripts, and the model provider. Verify each one at selection. Texas HB 300 likely reaches Rick directly as a covered entity; its exact intervals are not quoted here because they were not verified.

THE AGENT IS THE REPLACEABLE PART

Deterministic service
Runs with no model. Auditable, testable, in the specification. Survives any model change.

interface

Agent surface A, B
Operates the service through a clean interface. Bounded, logged, human-signed where it acts.

swappable

Model provider
Named vendor, own BAA, US residency. Swap it and nothing to the left changes. A locally hosted model is a specification-phase option.

WHY THIS SHAPE KEEPS THE CONTINUITY PROMISE HONEST

If Rick is gone, the center holds the specification, the center-specific code, its configuration and data, the contract terms for the engines, runbooks for every procedure, and vendor contracts in its own name. The services keep running with no agent at all. An integrator who has never met Rick can operate them from the specification, and can put a different agent, or no agent, on top. The one thing not promised anywhere in this set is payer-portal automation, which is the most fragile item in the inventory.

One environment, isolated instances. The owners decide whether to go further; the data boundary does not move.

The default is the industry standard, security first: one environment under one contract, each organization in its own isolated instance, drawn below. Never one shared database.

Every service on pages 1 to 4, per-organization data **never pooled** in either case

TIER 1, WRITTEN ONCE. ONE ENVIRONMENT, ONE CONTRACT.

One specification, reused for every center. One environment to operate, patch and monitor.

Order detection, packet assembly, deadline tracking, the records clock, the reporting joins, the report assistance path. Written once, with each organization's mappings and configuration filled in per site, inside one environment or in a separate stack.

FIRST ORGANIZATION

Its own isolated instance.

- Its own copy of every service: prior auth, reporting, backfill, records, report assistance.
- Its own database and its own encryption keys. Its own audit log.
- Its own credentials and role-based access, scoped to this organization alone.
- Its own model account under its own BAA. US residency verified for this instance.
- Its own principals approve its own sign-off rules; the four access terms go into whichever vendor contract applies.

HARD BOUNDARY, NO SHARED DATA

SECOND ORGANIZATION

Its own isolated instance.

- Its own copy of every service: prior auth, reporting, backfill, records, report assistance.
- Its own database and its own encryption keys. Its own audit log.
- Its own credentials and role-based access, scoped to this organization alone.
- Its own model account under its own BAA. US residency verified for this instance.
- Its own principals approve its own sign-off rules; the four access terms go into whichever vendor contract applies.

HARD BOUNDARY, NO SHARED DATA

THIRD ORGANIZATION

Its own isolated instance.

- Its own copy of every service: prior auth, reporting, backfill, records, report assistance.
- Its own database and its own encryption keys. Its own audit log.
- Its own credentials and role-based access, scoped to this organization alone.
- Its own model account under its own BAA. US residency verified for this instance.
- Its own principals approve its own sign-off rules; the four access terms go into whichever vendor contract applies.

THE DEFAULT: ONE ENVIRONMENT, ONE CONTRACT, ISOLATED INSTANCES

Security first, and the cost stays low.

The isolation lives at the data layer, where a regulator, an auditor, or a partner's lawyer expects to find it: separate databases, separate encryption keys, separate logins, separate audit logs. The environment is operated, patched and monitored once. One owner or one joint venture owning every center: this is the whole answer. A report that spans organizations is a later, explicit decision by all the principals, built read-only, never as a merged record.

THE VARIANT: SEPARATELY OWNED, FULLY SEPARATE HOSTING

A fully separate stack, at higher cost, as that owner's choice.

A separately owned center whose owner wants its own hosting and its own books gets a stack of its own: its own host, environment, contract and bills. It adds nothing for security that the isolated instance does not already give; it adds separation of books and billing, which is a legitimate reason and not a security argument. Either works. What is never built is one shared database with a center column.

WHAT IS SHARED IN BOTH CASES

The specification and the engines. Never a patient record.

- The specification, Tier 1, written once.
- Rick's engines, Tier 3, used by the center under the contract.
- The runbooks and the vendor evaluation criteria.
- In the default, the environment and its operations. Never a database, a key, a login, or an audit log.

Five phases, November to May, so everything works before your new center opens.

Target opening around June 2027, with about a month of buffer after go-live. Every date here is a target that holds only while the dependencies in the right-hand panel hold. If one slips, the dates move with it; the price does not.

Updated 2026-11-02
five-phase engagement

SEQUENCE. EACH PHASE STARTS ONLY AFTER THE PREVIOUS DELIVERABLE IS SIGNED OFF.

1. Discovery. Nov to mid-Dec 2026
Patient flow, services and modalities, volumes, payers, what is automated at opening. Deliverable: Requirements and Roadmap.

2. Vendor selection. Mid-Dec to Jan
Scheduling, RIS and PACS, billing, phones, prior auth. Demos and scoring. Deliverable: Vendor Selection Report, contracts to sign.

3. Design and staffing. Feb 2027
Systems, network, workflows, security and HIPAA, how each role's work runs. Deliverable: Architecture, Workflow and Staffing Plan.

4. Build and configure. Mar to Apr
Systems set up and connected, RadAssist configured, prior auth automated. Deliverable: all systems working in test.

5. Test, train, go-live. May 2027
Staff trained on the real systems. Deliverable: signed go-live readiness, plus opening-week support.

Opening. ~June 2027
About a month of buffer after go-live. If the opening date moves, every phase moves with it; the price does not.

OPERATIONS AGENT

Runs one workflow inside the center's systems.

Prior authorization, cancellation backfill, records requests. Works through the RIS (HL7 v2, DICOM) and the payers' own connections (FHIR prior authorization API, EDI), never by driving a portal screen. Assembles, checks, tracks, drafts. **A person on the center's staff approves anything that leaves the building.**

- Worked example: order arrives in the RIS, coverage and authorization rule checked, request assembled and checked against payer rules, staff approves, submitted electronically, decision clock tracked, approval scheduled, denial drafted as an appeal for a person to sign.

OWNER'S AGENT

Answers the owner, by text or phone. Reads, never acts.

Plain questions over the center's own operational data, answered from the reporting service, not from memory. A Monday report: volumes by modality, reader, and referrer; authorization and denial rates; reconciliation.

- **Calculated and checked:** totals and rates are computed by code and verified; if a figure cannot be verified the agent says so.
- Inside the FDA administrative exclusion: business analytics and practice management by name.
- No patient contact and no clinical surface.

DEPENDENCIES. THE DATES HOLD ONLY WHILE THESE HOLD.

Seven things have to be true before the build (Phase 4) can start.

- A signed agreement.
- Vendors selected and contracted in Phase 2, with sandbox access to their API, HL7, and DICOM interfaces.
- Test access to payer prior authorization connections, FHIR and EDI.
- HIPAA business associate agreements executed, including the model vendor's.
- One named owner contact who approves.
- Expected volumes and payer mix, before payer-specific design starts.
- Equipment delivery dates, so the build lands on the install.

VENDOR-NEUTRAL BY DESIGN

Works with whatever RIS and PACS the center picks.

If the PACS or RIS vendor ships its own version of a feature, the center can use it. The four access terms (API access, HL7 interface count and fees, export rights, DICOM query permissions) keep the center's data reachable whichever way that goes. Guardrails throughout: a person signs anything clinical or outbound; HIPAA BAAs; patient data stays in the United States.

What the demo really does today, and what replaces each simulated piece in production.

Live at **rfinlay.github.io/imaging-priorauth-demo**. Synthetic data only. Solid warm boxes are real today; dashed boxes are simulated or synthetic and labeled that way in the demo. The bottom row is the production path the build milestones deliver.

Built 2026-10-22
Script: **/imaging-priorauth-demo/script/**

THE DEMO, AS BUILT. LEFT TO RIGHT, ONE ORDER.

1. RIS order
SIMULATED
A synthetic HL7 v2.5.1 ORM^O01 message, generated per order and displayed. No live RIS connected.

2. Prior auth agent
REAL
Checks the auth rule, assembles a **Da Vinci PAS FHIR bundle** (Claim, ServiceRequest, QuestionnaireResponse, Patient, Coverage, more). A person approves it first.

3. HL7 Da Vinci PAS reference sandbox
REAL
POST Claim/$submit to prior-auth.davinci.hl7.org returns HTTP 201 and a tracking number; a **GET** reads the stored ClaimResponse back.

4. Payer decision
SIMULATED
The sandbox has no imaging rules, so approve or deny is scripted and labeled.

5. Appeal
DRAFTED, HUMAN SIGNS
Drafted from the denial and notes; the physician edits and signs.

6. Monday report
SYNTHETIC
Turnaround, denial, appeal, overturn rates; placeholder numbers.

HUMAN SIGN-OFF: packet approval before step 3 · appeal signature at step 5. The agent never submits or signs on its own. CMS clocks (7 days standard, 72 hours expedited) are real logic on a fast-forward demo clock.

THE PRODUCTION PATH. WHAT PHASE 4 (BUILD AND CONFIGURE) SWAPS IN.

Your RIS
The real RIS through its vendor's HL7 and API sandbox (Phase 4), then production. Orders arrive as HL7 v2 ORM; results back as ORU. Contract terms one and two.

Prior auth agent, in production
Same pipeline. The model vendor operates under a signed **BAA**, US data residency. Drafts, checks, and tracks; never screen-drives a payer portal. One instance per center, its own audit log.

Real payers
FHIR PAS where the payer runs it (Medicare Advantage, Medicaid, CHIP, exchange plans from 2027-01-01). **EDI 278** through the clearinghouse where not. A human-assisted path for fax and portal payers. Test access first (Phase 4).

Payer decision
Real ClaimResponse from the payer, on the CMS clock. The agent schedules on approval.

Appeal
Model draft under the BAA. The radiologist or ordering physician signs. Nothing outbound unsigned.

Monday report
Your own RIS and billing data, calculated and checked. The owner's agent answers questions by text or call.

HUMAN SIGN-OFF POINTS, PRODUCTION

A person stays in charge at every gate that leaves the building.

- Your coordinator approves each prior authorization packet before submission.
- A physician signs every appeal; the clinical argument is theirs.
- The radiologist signs every report (RadAssist, unchanged: calcium, thyroid, and spine all draft from his template and his rules; nothing interprets an image).
- The owner approves the sign-off rules for each center, and any change to them.
- Graduation from approve-every-item to sampled review happens only when the owner says so, in writing.

WHAT MAKES PRODUCTION POSSIBLE

The dependencies on page 7, mapped to this diagram.

- **RIS box:** RIS and PACS selected; HL7, API, and DICOM sandbox access granted; the four contract terms signed.
- **Agent box:** HIPAA BAAs executed with the model vendor and every vendor touching patient data; US residency verified.
- **Payer box:** payer FHIR and EDI test access; payer contracts and credentialing; payer mix known before payer-specific rules.
- **Report box:** real volumes after opening; until then, the report runs on test data.
- Vendor-neutral throughout: works with whatever RIS and PACS the center picks.